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Maneuver Coordination Using V2I to Improve Traffic Efficiency at Intersections

SKIM

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An edge server at a signal-free intersection schedules connected automated vehicles first come, first served, and tells each one when to cross, so most never have to stop.

PROBLEM Signalized and uncontrolled intersections force stops that waste travel time and raise emissions. Centralized reservation schemes exist, but many omit the message exchange a real deployment would need, and the authors aim to fill that gap.

METHOD Each approaching vehicle reports to a controller on a multi-access edge computing server next to the intersection. The controller assigns crossing slots in arrival order, which the authors call first-in-first-scheduled, and vehicles adjust speed to meet their slot, stopping only under heavy congestion. The paper also specifies the request and assignment message exchange between vehicle and controller.

RESULTS The abstract gives no numbers. It claims significant gains in travel time and emissions over uncontrolled intersections with and without traffic lights, and over one unnamed benchmark from the literature, across a range of traffic densities in simulation.

CAVEAT First-come scheduling at an intersection manager is essentially the reservation idea Dresner and Stone proposed nearly twenty years ago, so the novelty rests on the protocol details and the edge deployment. A reviewer would want to know how the scheme behaves when requests or assignments are lost or delayed, and what happens when some vehicles are not connected. The abstract shows no sign that either was tested. The evaluation is simulation only, against a single benchmark, and arrival-order scheduling is known to fall behind optimized ordering at high demand.

WHY IT MATTERS This is peripheral to your work. Conflicts are resolved by a central authority on the roadside, not negotiated between vehicles, and there are no proposals, acceptances or trajectory exchanges between peers. The one useful piece is the concrete vehicle-to-controller message sequence, which is a reference point if you need to contrast centralized and distributed coordination in a related-work section.

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